



Ekimetrics.

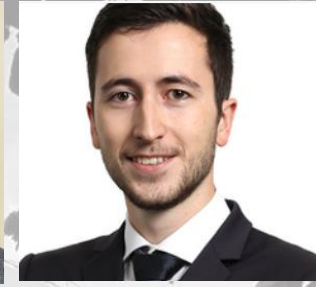
DSBA Hackathon

2026

PARIS | LONDON | NEW YORK | HONG KONG

Ekimetrics.
Data science for business

The ESSEC Squad



PARIS | LONDON | NEW YORK | HONG KONG

Ekimetrics in a nutshell

Ekimetrics.

**Leader in data science
and AI-powered
solutions for sustainable
business performance
with +350 clients and
+1,000 projects
worldwide.**

**WE ARE A PURPOSE-DRIVEN COMPANY
WITH SUSTAINABILITY AT THE CORE
OF OUR DNA.**



+17

YEARS
Experience in data science

+400

**DATA EXPERTS
& INDUSTRY
SPECIALISTS** Worldwide

+50

COUNTRIES
Where we lead projects



 **NEW YORK**

 **PARIS**

 **SHANGHAI**

 **LONDON**

 **HONG KONG**

Pending
B-Corp
By 2024

**+10 millions
tCO2e**

savings objective on
behalf of our clients by
2030

100% of employees
trained to
environmental issues

Ekimetrics.

Our mission and purpose

Data Science for sustainable business transformation

A good company

Delivering Data Science for business solutions **since 2006** with **400+ tech profiles** in 5 countries

A company that **commits to higher standards & responsibilities**, on the road towards BCORP in 2022 and Net Zero in 2025

One of the 30/150 precursor members of the “**Convention des Entreprises pour le Climat**”



Doing sustainable AI

100% of Eki employees **trained and aware on social & environmental challenges** (Climate Fresk, Climate trainings, Vendredi challenges).

Strongly against using technology for the sake of technology. We invest in R&D to develop unbiased and low-carbon footprint AI algorithms, and always rethink why we solve problems and how we can develop frugal tech solutions.

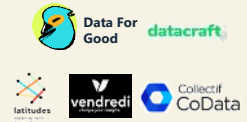
Open sourcing our best practices on our Eki.Lab website
<https://ekimetrics.github.io/>

For positive impact

Partnerships with sustainability & climate specialists



Active member of the “tech for good” ecosystem



AI for Sustainability solutions help businesses perform within the planetary boundaries and transition to a low-carbon economy

Build analytics capabilities and enhance your actions and decision-making power

DATA SCIENCE TRANSFORMATION

TRANSFORM AND GROW YOUR BUSINESS

Data Architecture & Governance

Data & Analytics Platform

Data Roadmap Management

MARKETING SCIENCE

ADVANCE YOUR BRAND AND BUSINESS STRATEGY

Live Mkt ROI

Digital Excellence

Brand Equity

Pricing Efficiency

CUSTOMER ANALYTICS

PUT THE CUSTOMER AT THE HEART OF BUSINESS

Tailored Experience

E2E Journey

Sentiment Analysis

Recommendation Engine

OPERATIONAL EXCELLENCE

RESHAPE YOUR E2E OPERATIONAL EFFICIENCY

E2E process automation & improvement

Stock optimisation

New products & services

AI FOR SUSTAINABILITY

ACT NOW TO BUILD YOUR BUSINESS WITHIN PLANETARY BOUNDARIES

Transition & Impact Score

AI Eco-Design

E2E sustainability footprint organisation

AN EKI OFFER
TO **ENHANCE**
YOUR ACTIONS
& **DECISION-MAKING**
POWER

Trusted by
+350 clients,
leaders in data
& sustainable
transformation

LUXURY



FINANCIAL SERVICES, BANKING & INSURANCE



TRANSPORTATION HOSPITALITY & QSR



MOBILITY



MEDIA & ENTERTAINMENT



RETAIL & ECOMMERCE



CONSUMER PACKAGED GOODS



ENERGY, UTILITIES & TELECOMS



HEALTH & PHARMA



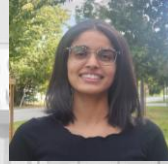
PARIS | LONDON | NEW YORK | HONG KONG

Kick off The Hackathon objectives

Ekimetrics.

The Hackathon Team

The entertainment team



Jury n°1



Jury n°2



Context

The goal is to extract KPIs from text documents while addressing two tasks:

- Injecting realistic hallucinations into extracted KPIs
- Detecting hallucinations that have been intentionally introduced

Participants will be divided into two groups:

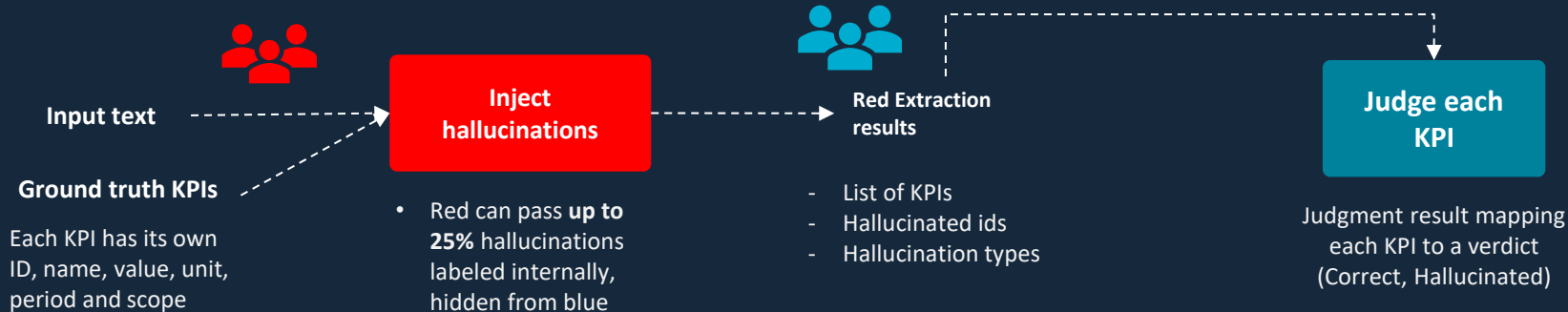
- **RED Team:** focused on creating plausible hallucinations
- **BLUE Team:** focused on identifying them

Throughout the day, teams will **alternate between both roles** to experience each task.

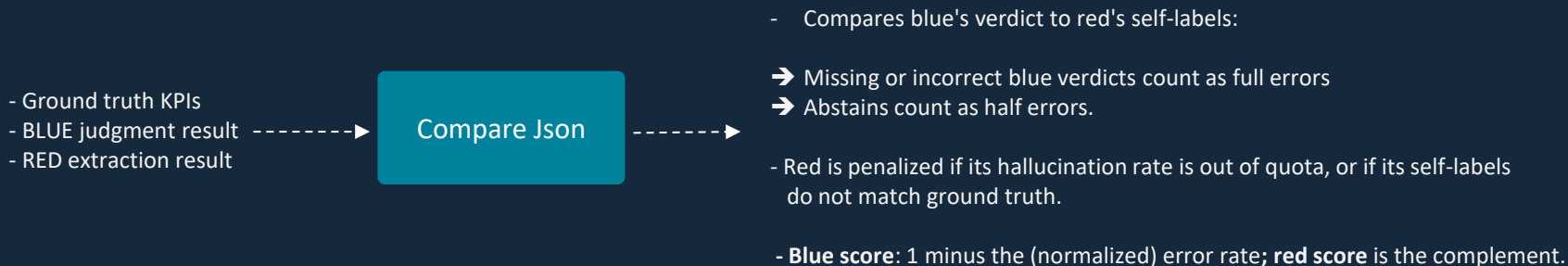
The challenge will cover **two types of documents:**

- Financial documents
- ESG (Environmental, Social & Governance) documents

Tournament



Deterministic Scorer



Deux règles à respecter dans la liste rendue

Mesurées par rapport aux **vrais KPIs du document** reçus en entrée

1

Reproduire au moins la moitié des vrais

Dans l'output, $\geq$ **la moitié** des KPIs réels reçus doivent être présents **tels quels** — sans la moindre modification.

$\geq 1/2$



des vrais KPIs, repris à l'identique

2

Ne pas trop halluciner

Le nombre de KPIs **faux / inventés** doit rester $<$ **un quart** du nombre de vrais KPIs reçus en entrée.

$< 1/4$



au plus 1 faux pour 4 vrais reçus

Une journée, deux mi-temps — les rôles s'inversent à midi

Le matin **Finance** · l'après-midi **ESG** — chaque groupe attaque (rouge) une fois, défend (bleu) une fois

Groupe A • 5 équipes



Siri



GLaDOS



R2-D2



Skynet



JARVIS



ROUGE

le matin · Finance

↓ À MIDI, ON ÉCHANGE



BLEU

l'après-midi · ESG

Finance Lite

Matin · Finance

Finance Long

Matin · Finance



Pause déjeuner

Midi

ESG Short

Après-midi · ESG

ESG Long

Après-midi · ESG

Groupe B • 5 équipes



Bender



Optimus



Alexa



Ultron



WALL-E

BLEU

le matin · Finance

↓ À MIDI, ON ÉCHANGE

ROUGE

l'après-midi · ESG

Finance Lite — 5 ROUGE × 5 BLEU = 25 duels

Chaque rouge produit **une seule liste** de KPIs · chaque bleu juge les **5 listes** reçues

ROUGE

1 liste KPI / équipe (vrais + hallucinations)



Siri

1 LISTE KPI

- id1 Revenue €4.2M ✓
- id2 EBITDA €0.9M ✓
- id3 Net margin 31% ⚠



GLaDOS

1 LISTE KPI

- id1 Revenue €4.2M ✓
- id2 Capex €0.3M ✓
- id3 Free cash flow €5.1M ⚠



R2-D2

1 LISTE KPI

- id1 Headcount 318 ✓
- id2 Revenue €4.2M ✓
- id3 EPS €2.40 ⚠



Skynet

1 LISTE KPI

- id1 EBITDA €0.9M ✓
- id2 Gross margin 47% ✓
- id3 Order backlog €12M ⚠



JARVIS

1 LISTE KPI

- id1 Revenue €4.2M ✓
- id2 Opex €3.1M ✓
- id3 ROE 28% ⚠

BLEU

juge les 5 listes · id par id



Bender

5 listes ● ● ● ● ●

✓ id1 correct ✗ id3 halluciné



Optimus

5 listes ● ● ● ● ●

✓ id2 correct ✗ id3 halluciné



Alexa

5 listes ● ● ● ● ●

✓ id1 correct ✗ id3 halluciné



Ultron

5 listes ● ● ● ● ●

✗ id3 halluciné ✓ id2 correct



WALL-E

5 listes ● ● ● ● ●

✓ id1 correct ✗ id3 halluciné

Comment on calcule les points

Le score mesure la performance d'un duel · le classement la transforme en points

LE BLEU · SCORE = MACRO-F1

F1 = moyenne harmonique de précision & rappel

VERDICT DU BLEU

	Halluciné	Correct
Faux	6 détecté ✓	1 raté X
Vrai	2 fausse alerte	9 ok ✓

CLASSE « HALLUCINÉ »

Précision = $6 / (6+2) = 0,75$

Rappel = $6 / (6+1) = 0,86$

F1 = $2 \cdot P \cdot R / (P+R) = 0,80$

Score Bleu = macro-F1 = 0,83

moyenne des 2 classes : $(0,80 \text{ « halluciné »} + 0,86 \text{ « correct »}) / 2$

LE ROUGE · JUGÉ EN MIROIR

Score Rouge = 1 – Score Bleu – pénalités quota

ici : $1 - 0,83 = 0,17$ (avant pénalités de quota)

Plus le Bleu se trompe, plus le Rouge marque — puis on retire les pénalités (slide quota).

Ce qui compte : le RANG

le score ne sert qu'à classer · chaque round distribue 5 → 1 pts

1		Optimus	F1 0,91	→	5 pts
2		WALL-E	F1 0,84	→	4 pts
3		Bender	F1 0,83	→	3 pts
4		Alexa	F1 0,72	→	2 pts
5		Ultron	F1 0,65	→	1 pts

Classement séparé Rouge et Bleu · égalité = points partagés (moyenne) · cumulé sur les 5 rounds → un **champion Rouge** et un **champion Bleu**.

Resources & Deliverables

- **ESG and Finance data already parsed and processed**
- **Access to a LLM API (with Groq)**
- **Code samples and explanations**

Deliverables

- **The code for the RED agent and BLUE agent**
- **A final presentation in front of a jury at the end of the day**

You will be assessed on the relevance of the proposed solution, the technical choices you made, and the quality of your final presentation

Code overview

Code QuickStart Guide

1

Data

ESG documents + financial documents

All documents are parsed, chunked, and processed and can be used directly.

2

Templates

A template for both RED and Blue team.

Each team has baseline ready to use agents for testing.

Defines types for extraction results, judgment result, examples to run a test match and the scoring method.

Final Presentation

At the end of the day, you'll be asked to present your solution. You'll be evaluated on 4 criteria



Presentation and communication

Clarity, Storytelling, Posture

5 points



Technical Soundness and Architecture

How does it work, Why does it work?

5 points



Innovation and Creativity

Going beyond obvious implementations; showing thoughtfulness

5 points



Use case and Insights

Identify real life challenges, use cases for the solution

5 points

Some useful links

- Building effective agents

[Building Effective AI Agents \ Anthropic](#)

- How we built our multi-agent research system

[How we built our multi-agent research system \ Anthropic](#)

- Demystifying eval for AI agents

[Demystifying evals for AI agents \ Anthropic](#)

- Groq API documentation

[API Reference - GroqDocs](#)